Proportional Hazards Regression

Research Opportunity Program (ROP)

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Introduction

- Today we will begin discussing regression models for time-to-event data
- There are a number of ways one could think about modeling the dependency between the time to an event and the factors that might affect it
- The two most common approaches are known as *proportional hazards models* and *accelerated failure time models*

Proportional Hazards

- We'll start with proportional hazards models
- As the name implies, the idea here is to model the hazard function directly:

$$\lambda_i(t) = \lambda(t) \exp(\mathbf{x}_i^T \boldsymbol{\beta})$$

- Here, the covariates act in a multiplicative manner upon the hazard function; note that the exponential function ensures that $\lambda_i(t)$ is always positive
- In this model, the hazard function for the ith subject always has the same general shape $\lambda(t)$, but can be, say, doubled or halved depending on a patient's risk factors

Exponential Regression

- In general, any hazard function can be used; today, we'll restrict attention to the constant hazard for the sake of simplicity
- Thus, the *exponential regression* model is:

$$\lambda_i(t) = \lambda \exp(\mathbf{x}_i^T \boldsymbol{\beta})$$

- Note that if $\mathbf{x}_i$ contains an intercept term, we will have a problem with identifiability – there is no way to distinguish β_0 and λ

Identifiability

- For a variety of reasons (convenience, simplicity, numerical stability, accuracy of approximate inferential procedures), it is preferable to estimate β_0 rather than λ, so this is the parameterization we will use
- Of course, having estimated β_0, one can easily obtain estimates and confidence intervals for λ through the transformation $\lambda = \exp(\beta_0)$
- In today's lecture notes, we will discuss how to estimate the regression coefficients and carry out inference concerning them, and then illustrate these results using the pbc data

Solving a Nonlinear System of Equations

- Maximum likelihood estimation of $\boldsymbol{\beta}$ is complicated in exponential regression by the need to solve a nonlinear system of equations
- This cannot be done in closed form; some sort of iterative procedure is required
- The basic idea is to construct a linear approximation to the nonlinear system of equations, solve for $\widehat{\boldsymbol{\beta}}$, re-approximate, and so on until convergence (this is known as the *Newton-Raphson algorithm*)
- We will begin by working out the score and Hessian with respect to the *linear predictor*, $\eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$

Log-likelihood, Score, and Hessian

- Under independent censoring and assuming $\widetilde{T}_i|\mathbf{x}_i \sim \mathrm{Exp}(\lambda_i)$, the log-likelihood contribution of the ith subject in exponential regression is

$$\ell_i(\eta_i) = d_i\eta_i - t_i e^{\eta_i}$$

- The score and Hessian are therefore

$$u_i(\eta_i) = d_i - t_i e^{\eta_i}$$
$$H_i(\eta_i) = -t_i e^{\eta_i}$$

Vector/Matrix Versions

- Letting $\boldsymbol{\mu}$ denote the vector with ith element $t_i e^{\eta_i}$ and $\mathbf{W}$ denote the diagonal matrix with ith diagonal element $t_i e^{\eta_i}$, we can rewrite the score and Hessian as

$$\begin{aligned}\mathbf{u}(\boldsymbol{\eta}) &= \mathbf{d} - \boldsymbol{\mu} \\ \mathbf{H}(\boldsymbol{\eta}) &= -\mathbf{W}\end{aligned}$$

- As we remarked earlier, solving for $\boldsymbol{\mu} = \mathbf{0}$ is complicated because $\boldsymbol{\mu}$ is a nonlinear function of $\boldsymbol{\eta}$; thus, consider the Taylor series approximation about $\tilde{\boldsymbol{\eta}}$

$$\begin{aligned}\mathbf{u}(\boldsymbol{\eta}) &\approx \mathbf{u}(\tilde{\boldsymbol{\eta}}) + \mathbf{H}(\tilde{\boldsymbol{\eta}})(\boldsymbol{\eta} - \tilde{\boldsymbol{\eta}}) \\ &= \mathbf{d} - \boldsymbol{\mu} + \mathbf{W}(\tilde{\boldsymbol{\eta}} - \boldsymbol{\eta})\end{aligned}$$

where $\boldsymbol{\mu}$ and $\mathbf{W}$ are fixed at $\tilde{\boldsymbol{\eta}}$

Solving for β

- So far, of course, we've ignored the fact that $\boldsymbol{\eta} = \mathbf{X}\boldsymbol{\beta}$ and that we're really estimating $\boldsymbol{\beta}$
- Substituting this expression into the previous equation and solving for $\boldsymbol{\beta}$, we obtain

$$\widehat{\boldsymbol{\beta}} \leftarrow (\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}\mathbf{X}^T(\mathbf{d} - \boldsymbol{\mu}) + \widetilde{\boldsymbol{\beta}}$$

- Again, this is an iterative process, which means that this is not an exact solution for $\widehat{\boldsymbol{\beta}}$; rather, we must solve for $\widehat{\boldsymbol{\beta}}$, recompute $\boldsymbol{\mu}$ and $\mathbf{W}$, re-solve for $\widehat{\boldsymbol{\beta}}$, and so on
- The Newton-Raphson algorithm will converge to the MLE (although this is not absolutely guaranteed) provided that the likelihood is log-concave and coercive, both of which (typically) hold for exponential regression

Crude R Code

- Below is some crude R code providing an implementation of this algorithm

```
b <- rep(0, ncol(X))
for (i in 1:20) {
  eta <- as.numeric(X%*%b)
  mu <- t*exp(eta)
  W <- diag(t*exp(eta))
  b <- solve(t(X) %*% W %*% X) %*% t(X) %*% (d-mu) + b
}
```

- This is crude in the sense that it isn't as efficient as it could be and in that it assumes convergence will occur in 20 iterations; a better algorithm would check for convergence by examining whether $\widehat{\beta}$ has stopped changing

Wald Approach

- Since $\widehat{\boldsymbol{\beta}}$ is the MLE, our derivation of the Wald results from earlier means that

$$\widehat{\boldsymbol{\beta}} \dot{\sim} \mathrm{N}\left(\boldsymbol{\beta}, \mathbf{I}^{-1}\right);$$

  we just have to work out the information matrix with respect to $\boldsymbol{\beta}$
- Applying the chain rule, we have

$$\widehat{\boldsymbol{\beta}} \dot{\sim} \mathrm{N}\left(\boldsymbol{\beta}, (\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}\right)$$

- It is very easy, therefore, to construct confidence intervals for β_j with $\widehat{\beta}_j \pm z_{1-\alpha/2}\mathrm{SE}_j$, where $\mathrm{SE}_j = \sqrt{(\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}_{jj}}$

Likelihood Ratio Approach

- The likelihood ratio approach, while desirable, is somewhat complicated in multiparameter settings where we lack closed-form estimates
- Consider the problem of obtaining a likelihood ratio confidence interval for β_j
- If β_j was the only parameter, this is simply a root-finding problem in which we determine the values β_L and β_U where $2(\ell(\widehat{\beta}_j) - \ell(\beta_j)) = \chi^2_{1,.95}$

Interpretation of Coefficients

- As in other regression models, the interpretation of the regression coefficients involves the effect of changing one factor while all others remain the same
- Consider a hypothetical comparison between two individuals whose explanatory variables are the same, except for variable j, where it differs by $\delta_j = x_{1j} - x_{2j}$:

$$\frac{\lambda_1(t)}{\lambda_2(t)} = \exp(\delta_j \beta_j)$$

Hazard Ratios

- Note that for any proportional hazard model, $\lambda_1(t)/\lambda_2(t)$ is a constant with respect to time
- This constant is known as the *hazard ratio*, and typically abbreviated HR, although some authors refer to it as the *relative risk* (RR)
- Thus, the interpretation of a regression coefficient in a proportional hazards model is that $e^{\delta\beta}$ is the hazard ratio for a δ-unit change in that covariate
- In particular, $\mathrm{HR} = e^{\beta}$ for a one-unit change
- So, for stage in our pbc example, $\mathrm{HR} = e^{0.564} = 1.76$; a one-unit change in stage increases a patient's hazard by 76%

The End.